Section 4.3 — Properties of Logarithms — Answer Key

Math 103: College Algebra

You may wish to reference this list of log rules while you work on this assignment. You should plan to memorize these in the long run.

$$\log_b(xy) = \log_b x + \log_b y \qquad \log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y \qquad \log_b(x^p) = p\log_b x \qquad \log_b x = \frac{\log_c x}{\log_c b}$$

Rewriting as Simple Logarithms

Rewrite each expression using the simplest possible logarithms.

1. $\log_4 s + 4log_4 z$
2. $2\log_{12} v - \log_{12} x$
3. $3\log_3 z + 2\log_3 v - 3\log_3 x$
4. $6\log_8 m - 3\log_8 n - \frac{1}{3}\log_8 w$
5. $\log_9 s + 4\log_9(s-7) - 2\log_9(s-2)$
6. $\log x + \frac{1}{2}\log(x-4) - \frac{1}{3}(2x+1)$

Rewriting as a Single Logarithm

Rewrite each expression as a single logarithm.

7. $\log_4(ms^4)$
8. $\log_7\left(\frac{v^6}{x}\right)$
9. $\log_8\left(\frac{w^5}{t^6x^3}\right)$
10. $\log_4\left(\frac{x^2\sqrt[3]{z}}{\sqrt[3]{y^2}}\right)$

Rewriting with change of base

Rewrite each expression in terms of the common logarithm.

11. $\log_5 x = \frac{\log x}{\log 5}$
12. $\log_3 4 = \frac{\log 4}{\log 3}$
13. $\log_{1/2}(x-3) = \frac{\log(x-3)}{\log\frac{1}{2}}$
14. $\log_8(x^2+1) = \frac{\log(x^2+1)}{\log 8}$

For each expression below:

a. Rewrite the expression as a single logarithm.

b. Evaluate the expression on your calculator.

c. Check by rewriting the single logarithm in exponential form.

15. $\frac{\log 4}{\log 2} = \log_2 4 = 2$
    $2^2 = 4$
16. $\frac{\log 25}{\log\frac{1}{5}} = \log_{1/5} 25 = -2$
    $\left(\frac{1}{5}\right)^{-2} = 25$
17. $\frac{\log 2}{\log 16} = \log_{16} 2 = \frac{1}{4}$
    $16^{1/4} = 2$
18. $\frac{\log 0.25}{\log 0.5} = \log_{0.5} 0.25 = 2$
    $(0.5)^2 = 0.25$

For each expression below:

a. Rewrite the expression in terms of the common logarithm.

b. Evaluate the expression on your calculator. Round your answer to two decimal places.

c. Check by rewriting the single logarithm in exponential form and evaluating on your calculator.

19. $\log_4(83) = \dfrac{\log 83}{\log 4} \approx 3.19$
$4^{3.19} \approx 83.28 \approx 83$

20. $\log_{1.05}(2.25) = \dfrac{\log 2.25}{\log 1.05} \approx 16.62$
$(1.05)^{16.62} \approx 2.25$

21. $\log_{0.7}(5) = \dfrac{\log 5}{\log 0.7} \approx -4.12$
$(0.7)^{-4.12} \approx 5.00$

22. $\log_{0.95}(0.31) = \dfrac{\log 0.31}{\log 0.95} \approx 22.83$
$(0.95)^{22.83} \approx 0.31$